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Cell processing system and method with preliminary process evaluation

Abstract

A cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive the at least one process parameter, to evaluate the process using the at least one process parameter before performing the process, and to carry out one or more actions based on the evaluation, such as providing an output estimate to the operator, preventing the process from being performed according to a comparison between a calculated condition and a control, or providing an error indication to the operator according to the calculated condition and a measured in-process condition.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3877634	12/1974	Rohde	N/A	N/A
4835372	12/1988	Gombrich	N/A	N/A
5053121	12/1990	Schoendorfer	N/A	N/A
5194145	12/1992	Schoendorfer	N/A	N/A
5240856	12/1992	Goffe	N/A	N/A
5458566	12/1994	Herrig	N/A	N/A
5536475	12/1995	Moubayed	N/A	N/A
5573678	12/1995	Brown	N/A	N/A
5769811	12/1997	Stacey	N/A	N/A
5865718	12/1998	Chan	N/A	N/A
5956023	12/1998	Lyle	N/A	N/A
6060022	12/1999	Pang	N/A	N/A
6175420	12/2000	Barry	N/A	N/A
6284142	12/2000	Muller	N/A	N/A
6325775	12/2000	Thom	N/A	N/A
6466879	12/2001	Cantu	N/A	N/A
6605223	12/2002	Jorgensen	N/A	N/A
6622052	12/2002	Rosiello	N/A	N/A
6716151	12/2003	Panzani	N/A	N/A
6730054	12/2003	Pierce	N/A	N/A
6736788	12/2003	Mongomery	N/A	N/A
7044927	12/2005	Mueller	N/A	N/A
7072769	12/2005	Fletcher-Haynes	N/A	N/A
7363167	12/2007	Csore	N/A	N/A

7390484	12/2007	Fraser	N/A	N/A
7430478	12/2007	Fletcher-Haynes	N/A	N/A
7651474	12/2009	Van Waeg	N/A	N/A
7963901	12/2010	Langley	N/A	N/A
8150548	12/2011	Raghibizadeh	N/A	N/A
8357298	12/2012	Demers	N/A	N/A
8415145	12/2012	Fukuda	N/A	N/A
8539573	12/2012	Newlin	N/A	N/A
8883499	12/2013	Hedrick	N/A	N/A
8900172	12/2013	Pohlmeier	N/A	N/A
8905959	12/2013	Basaglia	N/A	N/A
8945376	12/2014	Cordisco	N/A	N/A
2002/0128584	12/2001	Brown	604/6.02	A61M 1/308
2002/0179544	12/2001	Johnson	N/A	N/A
2003/0018289	12/2002	Ng	N/A	N/A
2003/0069480	12/2002	Ng	N/A	N/A
2003/0199379	12/2002	Schneider	N/A	N/A
2004/0248077	12/2003	Rodriguez Rilo	N/A	N/A
2005/0070837	12/2004	Ferrarini	N/A	N/A
2005/0215937	12/2004	Spinale	N/A	N/A
2007/0219826	12/2006	Brodsky	N/A	N/A
2008/0040153	12/2007	Davis, Jr.	N/A	N/A
2009/0191174	12/2008	Boudreau	N/A	N/A
2009/0211987	12/2008	Min	N/A	N/A
2009/0215602	12/2008	Min	N/A	N/A
2011/0045959	12/2010	Kurihara	N/A	N/A
2011/0206643	12/2010	Fulga	N/A	N/A
2011/0244443	12/2010	Van Rijn	N/A	N/A
2013/0092630	12/2012	Wegener	210/90	A61M 1/265
2013/0267884	12/2012	Boggs	N/A	N/A
2013/0299399	12/2012	Suffritti	N/A	N/A
2013/0334139	12/2012	Blickhan	N/A	N/A
2014/0081193	12/2013	Watters	N/A	N/A
2014/0234183	12/2013	Kolenbrander	N/A	N/A
2014/0266983	12/2013	Christensen	N/A	N/A
2014/0282181	12/2013	Declerck	N/A	N/A
2014/0291248	12/2013	Foley	N/A	N/A
2017/0262601	12/2016	Binninger	N/A	N/A
2018/0015418	12/2017	Binninger	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20305506	12/2003	DE	N/A
1011752	12/2003	EP	N/A
2720730	12/2013	EP	N/A
2745864	12/2013	EP	N/A
8102979	12/1980	WO	N/A
0020053	12/1999	WO	N/A
02062484	12/2001	WO	N/A
02069793	12/2001	WO	N/A

02088897	12/2001	WO	N/A
02089340	12/2001	WO	N/A
2004032999	12/2003	WO	N/A
2009072510	12/2008	WO	N/A
2012021167	12/2011	WO	N/A
2012125457	12/2011	WO	N/A
2012125470	12/2011	WO	N/A
2013043433	12/2012	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report for EP 21190537.7 dated Dec. 21, 2021. cited by applicant
European Patent Office, extended European Search Report, counterpart EP Appl. No. 17160556.1, dated Oct. 19, 2017 (10 pages). cited by applicant
Fresenius Kabi, Lovo Cell processing system, Filtration Technology Designed for Labs Like Yours. (2014). cited by applicant
Fresenius Kabi, Lovo Cell Processing System, Choose filtered. (2014). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/398,780, filed Aug. 10, 2021, which is a continuation of U.S. application Ser. No. 15/456,853, filed Mar. 13, 2017, now U.S. Pat. No. 11,191,879, which claims the benefit of U.S. Provisional Application No. 62/342,675, filed May 27, 2016, which are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure is generally directed to systems and methods for evaluating a procedure or process, or parts or portions thereof, prior to the process being performed, and to biological fluid processing systems and methods employing such. More particularly, the present disclosure is directed to the controlled processing of biological fluid using a disposable fluid circuit and a reusable processing machine, where the processing is evaluated before it is performed.

BACKGROUND

(2) The processing of biological fluid such as blood or blood components typically involves using a reusable processing machine (“hardware”) and a disposable fluid circuit adapted for mounting or other association with the reusable apparatus. The fluid circuit typically includes (plastic) bags and associated tubing that defines a flow path through the circuit. The disposable fluid circuit may also include one or more separation devices where the biological fluid/cells can be separated into two or more components, washed or otherwise processed. Separation devices may separate the biological fluid based on centrifugal separation and/or, as described below, membrane separation.

(3) Conventionally, assumptions are made regarding the operation of the apparatus, for example, concerning the time required to perform a procedure and the amount of fluid generated as waste as a consequence. Often, these assumptions are based on empirical data derived under conditions similar to, but not identical to, the conditions under which the biological fluid will be processed presently. As such, the operator is left to adjust his or her assumptions based on his or her experience with the apparatus and the process being performed. This can prove problematic where

little or no empirical data exists on which the operator can base his or her assumptions, or where the operator has only limited experience with the apparatus, the process being performed, or the empirical data.

SUMMARY

(4) In one aspect, a blood processing system includes a blood processing apparatus. The blood processing apparatus includes reusable hardware and a disposable fluid circuit. The blood processing apparatus includes a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component in a product container, the reusable hardware including a weight scale configured to weigh the product container and an air detector sensor. The blood processing system also includes a touch screen configured to receive input from an operator, a transmitter/receiver device configured to communicate over a network and to receive process parameters from a server computer, and a processing circuit coupled to the blood processing apparatus and the network, the processing circuit configured to receive the process parameters from the server computer and store them in a memory, the processing circuit configured to store default process parameters in the memory. The processing circuit is configured to receive an activation from the operator to switch on the blood processing system, conduct self-checks on the blood processing apparatus, receive input data for a value of a process parameter from the touch screen, verify that the value is within a preset range for the value. The processing circuit may be configured to, prior to operating the blood processing apparatus according to a centrifugal separation process, store a pre-process calculation of a volume of the blood component to be collected in the product container and provide an indication of the volume to an operator via the touch screen, and/or store a pre-process calculation of a volume of the blood component to be collected in the product container based at least in part on the value of the process parameter received from the touch screen. The processing circuit is further configured to prompt the operator to mount the disposable fluid circuit and automatically check to determine whether the disposable fluid circuit is installed. In a further aspect, the processing circuit may be configured to prime the tubing of the disposable fluid circuit with a fluid, commence the centrifugal separation process on the biological fluid, check the air detector sensor for a fluid/air interface, weigh the product container during the centrifugal separation process, and process the biological fluid until the volume of the blood component to be collected is collected. The processing circuit may be further configured to prompt the operator upon completion of the centrifugal separation process.

(5) In another aspect, a blood processing system includes a blood processing apparatus. The blood processing apparatus includes reusable hardware and a disposable fluid circuit, the blood processing apparatus further includes a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component to a product container, the reusable hardware including a weight scale configured to weigh the product container and an air detector sensor. The blood processing system includes a touch screen configured to receive input from an operator and a control unit coupled to the blood processing apparatus, the control unit configured to store default process parameters in the memory, the control unit configured to receive an instruction from the operator to activate the blood processing system, conduct self-checks on the blood processing apparatus, receive input data for a value of a process parameter from the touch screen and, prior to operating the blood processing apparatus according to a centrifugal separation process, calculate a volume of the blood component to be collected in the product container based at least in part on the value of the process parameter received from the touch screen. The control unit is configured to provide an indication of the volume to be collected to an operator via the touch screen, prime tubing of the disposable fluid circuit with a fluid, commence the centrifugal separation process on the biological fluid using the default process parameters, check the air detector sensor for a fluid/air interface, weigh the product container during the centrifugal separation process and process the biological fluid to collect the volume of the blood component to be collected.

(6) In another aspect, a cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive the at least one process parameter via the input, to calculate at least one output based on the at least one process parameter, and to provide the at least one calculated output to an operator prior to causing the processor to operate according to the process.

(7) In another aspect, a method for evaluating a process is provided, the process to be performed using a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams. The method includes receiving at least one process parameter to be used during the process, calculating at least one output based on the at least one process parameter, and providing the at least one calculated output to an operator prior to operating the processor according to the process.

(8) In a further aspect, a cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive the at least one process parameter via the input, to calculate at least one in-process condition based on the at least one process parameter, to compare the at least one calculated in-process condition with at least one in-process condition control, and to provide an error indication to an operator if the at least one calculated in-process condition does not match the at least one in-process condition control prior to causing the processor to operate according to the process.

(9) In another aspect, a method for evaluating a process is provided, the process to be performed using a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams. The method includes receiving at least one process parameter to be used during the process, calculating at least one in-process condition based on the at least one process parameter, comparing the at least one calculated in-process condition with at least one in-process condition control, and providing an error indication to an operator if the at least one calculated in-process condition does not match the at least one in-process condition control prior to operating the processor according to the process.

(10) In yet another aspect, a cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive the at least one process parameter via the input, to calculate at least one in-process condition based on the at least one process parameter prior to causing the processor to operate according to the process, to subsequently cause the processor to operate according to the process, to measure at least one in-process condition during operation of the processor, to compare the at least one calculated in-process condition with at least one measured in-process condition, and to provide an error indication to an operator if the at least one measured in-process condition does not match the at least one calculated in-process condition.

(11) In a still further aspect, a method for evaluating a process is provided, the process to be performed using a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams. The method includes receiving at least one process parameter to be used during the process, calculating at least one in-process condition based on the at least one process parameter prior to operating the processor according to the process, causing subsequently

the processor to operate according to the process, measuring at least one in-process condition during operation of the processor, comparing the at least one calculated in-process condition with the at least one measured in-process condition, and providing an error indication to an operator if the at least one measured in-process condition does not match the at least one calculated in-process condition.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The disclosure will be more fully understood from the following description taken in conjunction with the accompanying drawings. Some of the figures may have been simplified by the omission of selected elements for the purpose of more clearly showing other elements. Such omissions of elements in some figures are not necessarily indicative of the presence or absence of particular elements in any of the exemplary embodiments, except as may be explicitly delineated in the corresponding written description. None of the drawings is necessarily to scale.
- (2) FIG. 1 is a frontal view of a reusable cell processing apparatus with a disposable fluid circuit loaded thereon.
- (3) FIG. 2 is a schematic view of the control circuitry of the apparatus of FIG. 1.
- (4) FIG. 3 is a schematic view of one embodiment of a disposable fluid circuit useful in the systems and methods described herein;
- (5) FIG. 4 is a frontal view of the reusable cell processing apparatus;
- (6) FIG. 5 is a perspective view of a separation/washing device using a spinning membrane;
- (7) FIG. 6 is a flowchart of one embodiment of a method of operating a reusable cell processing apparatus with a disposable fluid circuit loaded thereon, such as is illustrated in FIG. 1, to process a biological fluid; and
- (8) FIG. 7 is a flowchart of one embodiment of a method for evaluating a procedure, or a portion thereof, to be performed on, for example, a reusable cell processing apparatus with a disposable fluid circuit prior to the procedure being performed using the apparatus and fluid circuit.

DETAILED DESCRIPTION

- (9) As illustrated in FIGS. 1 and 2, a cell processing system includes a processor **100, 200** to receive a biological fluid to be processed, a control unit (or controller) **300** coupled to the processor, the controller **300** configured to operate the processor **100, 200** according to a procedure or process. The controller **300** is also configured to evaluate the process, or parts or portions thereof, prior to operating the processor **100, 200** according to the process. In particular, the controller may perform a pre-process calculation of one or more outputs (e.g., time to conduct the process, or parts or portions thereof, volume of wash media consumed, volume of waste fluid generated, volume of residuals remaining in final product, etc.) and/or one or more in-process conditions. The calculation may involve a mathematical model of the entire process, from initiation to completion. The calculated outputs and/or in-process conditions may be provided or displayed to the operator so that the operator may compare them against the operator's existing assumptions or to provide the operator with estimates that can be used on a going-forward basis. These outputs and/or in-process conditions may be compared with control values to warn the operator, prior to performing the process, that the process might exceed the abilities or physical constraints on the processor **100, 200** if performed. Further, the outputs and/or in-process conditions may be compared with values measured during operation of the processor **100, 200** to warn the operator that the process, as performed, is generating outputs or in-process conditions that differ from those expected, pre-process.
- (10) As explained in detail below, the processor **100, 200** may include a disposable fluid circuit **100** (see also FIGS. 3 and 5) and reusable hardware **200** (see also FIG. 4). According to the illustrated

embodiments, the disposable fluid circuit **100** may include a spinning membrane **101**, at least one container **102**, **122**, **135a**, **135b**, **140**, **150** (of which at least containers **102**, **135a**, **135b** may be initially separate and then connected to the remainder of the circuit **100** at the time of processing), and tubing **106**, **120**, **128**, **132a**, **132b**, **162**, **166**, **168** connecting the spinning membrane **101** and the one or more containers **102**, **122**, **135a**, **135b**, **140**, **150**. As is also illustrated, the reusable hardware **200** may include at least one drive **248** to spin the spinning membrane **101**, at least one scale **240**, **242**, **244**, **246** to weigh the at least container **102**, **122**, **140**, **150** and contents thereof, and at least one pump **202**, **204**, **206** to receive the tubing **162**, **166**, **168** and pump fluid therethrough such as by peristaltic action, although other types of pumps and pumping action may be used.

(11) The controller **300** may, according to the embodiments, include a programmable microprocessor **304**, which microprocessor **304** may be coupled to the at least one input **302** and may be programmed to operate the processor according to a process. In particular, the controller may be programmed to carry out a pre-process evaluation, resulting in the calculation of one or more outputs and/or one or more in-process conditions. As mentioned above, these outputs (and/or in-process conditions) may be provided to the operator, pre-process, for comparison against the operator's existing assumptions, for example. These outputs and/or in-process conditions may be compared with control values, pre-process, or values measured during operation of the processor, to provide warnings or error indications to the operator or limit or prevent the operation of the processor according to the process.

(12) In addition, the embodiments illustrate a method of operating a cell processing system, the cell processing system including a processor **100**, **200** to receive a biological fluid to be processed. The method may include a pre-process evaluation, resulting in the calculation of one or more outputs and/or one or more in-process conditions. These outputs (and/or in-process conditions) may be provided to the operator, pre-process, for comparison against the operator's existing assumptions, for example. Alternatively, these outputs and/or in-process conditions may be compared with control values, pre-process, or values measured during operation of the processor, to provide warnings or error indications to the operator to limit or prevent the operation of the processor according to the process.

(13) An embodiment of the afore-mentioned system and method thus may provide one or more of the following advantages. Initially, the system and method may provide estimates for outputs that are uniform, considering that the controller is providing the estimates based on a uniform evaluation method and system. Moreover, the system and method may prevent operation of the processor in a manner that will damage the source material, create unsafe operating conditions, and/or produce a final product that does not meet the desired specifications. Further, the system and method may control the operation of the processor in a manner that will limit damage to the processor, the product, or both. In addition, the system and method may alert the operator to conditions during the operation that have deviated from the expected conditions, and that might require user intervention (e.g., the attachment of additional wash media containers or containers with greater capacity) to prevent interruption of processing. Other advantages may also result.

(14) While the foregoing discussion references an embodiment in the form of a cell processing system, other systems may incorporate this technology as well. These systems may share the technical challenges faced by the aforementioned cell processing system, and incorporation of the technology may provide similar advantages.

(15) For example, a separation system, more particularly a filtration system, or even more particularly a microfiltration system, also may include a processor to receive a fluid to be processed and a controller. Further, certain embodiments of such a processor may include a disposable fluid circuit (which circuit may include a membrane used for filtration) and reusable hardware, and the controller may be configured to operate the processor. According to such a system, the controller may be configured to evaluate the process, or parts or portions thereof, prior to operating the

processor according to the process. This evaluation may include performing a pre-process calculation of a mathematical model of the entire process, from initiation to completion. The evaluation may also result in calculated outputs and/or in-process conditions, which may be provided to the operator or which may be used to generate automated warnings before or during the process that are provided to the operator.

(16) Having thus described the system and method in general terms, the details of the system and method are described in detail.

(17) As mentioned above, the systems disclosed herein typically include a reusable separation apparatus and one or more disposable processing circuits adapted for association with the reusable apparatus, which apparatus and circuit(s) define the processor. The reusable separation apparatus may be any apparatus that can provide for the automated processing of biological fluid. “Biological fluid” includes without limitation blood and blood components, and “cell” or “biological cell” includes without limitation blood cells, such as red cells, white cells and platelets. By “automated,” it is meant that the apparatus can be programmed to carry out the processing steps of a biological fluid processing method without substantial operator involvement. Of course, even in the automated system of the present disclosure, it will be understood that operator activity may be involved, including the loading of the disposable fluid circuits and entering processing parameters. Additional manual steps may be required as well. However, the reusable apparatus can process biological fluid through the disposable circuit(s) described below without substantial operator intervention.

(18) The illustrated processing apparatus is typically capable of effecting the separation of a biological fluid that includes biological cells into two or more components or fractions. Thus, the reusable apparatus may generate conditions that allow for the separation of a biological fluid into selected components or fractions. One preferred machine for separating biological fluid into its constituent components or fractions uses a spinning porous membrane. An example of such machine is the Autopheresis C® sold by Fenwal, Inc. of Lake Zurich, Illinois, which is an affiliate of Fresenius Kabi AG of Bad Homburg, Germany. A detailed description of a spinning membrane may be found in U.S. Pat. No. 5,194,145 to Schoendorfer, which is incorporated by reference herein in its entirety, and in International (PCT) Application No. PCT/US2012/028492, filed Mar. 9, 2012, the contents of which are also incorporated herein in their entirety. In addition, systems and methods that utilize a spinning porous membrane are also disclosed in U.S. Provisional Patent Application No. 61/537,856, filed on Sep. 22, 2011, and International (PCT) Application No. PCT/US2012/028522, filed Mar. 9, 2012, the contents of each are incorporated herein by reference. The references identified above describe a membrane-covered spinner having an interior collection system disposed within a stationary shell. While a detailed discussion of the separation device is beyond the scope of this application, the spinning membrane separation device is shown in FIGS. 6, 7(a)-7(b) of the reference cited and is discussed below in general terms. In another embodiment, the reusable apparatus may generate a centrifugal field to effect separation.

(19) Turning now to FIG. 3, the systems described herein include at least one disposable fluid circuit **100** for use in the processing of biological fluid. While the circuits described herein may be used as stand-alone circuits, more preferably, at least two or more disposable fluid circuits are used in combination and in series for the separation, washing, volume reduction and/or other processing of a biological fluid. Circuit **100** may include an integrated separation device, such as, but not limited to, the spinning membrane **101** described above. Circuit **100** may also include waste container **140**, product container **150**, and in-process container **122**. Disposable fluid circuits of the type described below may further include sampling assemblies (not shown) for collecting samples of source biological fluid, “final” product, or other intermediate products obtained during the biological fluid processing.

(20) As will be seen in the Figures and described in detail below, the disposable fluid processing circuits include tubing that defines flow paths throughout the circuits, as well as access sites for

sterile or other connection to containers of processing solutions, such as wash solutions, treating agents, or sources of biological fluid. As shown in FIG. 3, the tubing of circuit **100** includes spaced tubing segments identified by reference numerals **162**, **166**, **168**. The tubing segments are provided for mating engagement with the peristaltic pumps **202**, **204**, **206** of the reusable hardware apparatus **200** discussed below. The containers and the plastic tubing are made of conventional medical grade plastic that can be sterilized by sterilization techniques commonly used in the medical field such as, but not limited to, radiation or autoclaving. Plastic materials useful in the manufacture of containers and of the tubing in the circuits disclosed herein include plasticized poly(vinyl chloride). Other useful materials include acrylics. In addition, certain polyolefins may also be used.

(21) As will be apparent from the disclosure herein, source containers may be attached in sterile fashion to the circuit **100**. Source containers **102** for connection to one disposable circuit may be the product containers **150** of another circuit used in an earlier step of the overall method of processing. Alternatively, the contents of a product container **150** may be further processed or separated and then transferred in sterile fashion to the source container **102** of a later-in-series fluid circuit.

(22) The biological cell suspension to be washed or otherwise treated is typically provided in a source container **102**, shown in FIG. 3 as (initially) not connected to the disposable set. As noted above, source container **102** may be attached (in sterile fashion) at the time of use. Source container **102** has one or more access sites **103**, **105**, one of which may be adapted for (sterile) connection to fluid circuit **100** at docking site **104**. Preferably, source containers may be attached in a sterile manner by employing sterile docking devices, such as the BioWelder, available from Sartorius AG, or the SCD IIB Tubing Welder, available from Terumo Medical Corporation. A second access port **105** may also be provided for extracting fluid from the source container **102**.

(23) As further shown in FIG. 3, tubing segment **106** extends from docking site **104** and is connected to further downstream branched-connector **118**. Branched-connector **118** communicates with tubing **106** and tubing **120**, which provides a fluid flow path from “in-process” container **122**, described in detail below. Tubing segment **124** extends from branched-connector **118** and is joined to a port of further downstream branched-connector **126**. A separate flow path defined by tubing **128** is also connected to a port of branched-connector **126**.

(24) In accordance with the fluid circuit of FIG. 3, one or more containers of wash or other processing/treating solution may be attached (or pre-attached) to set **100**. As shown in FIG. 3, tubings **132a**, **132b** (defining a flow path) preferably include and terminate in an access site such as spike connectors **134a**, **134b**. Access sites **134a**, **134b** are provided to establish flow communication with containers **135a**, **135b** (shown in FIG. 1) of a wash fluid, such as saline or other solution. Tubings **132a**, **132b** may include in-line sterile barrier filters **130a**, **130b** for filtering any particulate from a fluid before it enters the flow path leading to second branched-connector **126** and, ultimately separator **101**. In one embodiment, the sterile barrier filters **130a**, **130b** may be 0.2 μm filters. The wash medium or fluid flows from the wash fluid source through tubing segments **132a**, **132b** where it is filtered by the sterile barrier filters **130a**, **130b** described above, and then passes through tubing **128** to the input of the branched-connector **126** described above.

(25) Tubing segment **136** defines a flow path connected at one end to branched-connector **126** and to an inlet port **20** of the separator **101**. Preferably, in accordance with the present disclosure, separation device **101** is a spinning membrane separator of the type described in U.S. Pat. Nos. 5,194,145 and 5,053,121, which are incorporated by reference, U.S. Provisional Patent Application Ser. No. 61/451,903 and PCT/US2012/028522, also previously incorporated herein by reference.

(26) As shown in FIG. 3 (and described in detail in connection with FIG. 5), the spinning membrane separator **101** has at least two outlet ports. Outlet **46** of separator **101** receives the waste from the wash (i.e., the diluted suspension medium) and is connected to tubing **138**, which defines a flow path to waste product container **140**. The waste product container **140** includes a further connection port **141** for sampling or withdrawing the waste from within the product container.

(27) Separation device **101** preferably includes a second outlet **48** that is connected to tubing segment **142** for directing the desired biological cell/fluid product to the in-process container(s) **122** or the product container **150**. To permit this, the other end of tubing segment **142** is connected to branched-connector **144**, which branches into and defines a flow path to one or more in-process containers **122** and a flow path to a “final” product container **150**. The product container **150** may also include a sampling assembly (not shown).

(28) FIG. **4** shows the front panel **201** of reusable hardware processing apparatus **200**, also referred to herein as “hardware”. Apparatus **200** may be of compact size suitable for placement on a table top of a lab bench and adapted for easy transport. Alternatively, apparatus **200** may be supported by a pedestal that can be wheeled to its desired location. In any event, as shown in FIG. **4**, apparatus **200** includes a plurality of peristaltic pumps such as pumps **202**, **204** and **206** on front panel **201**. Pump segments of the disposable fluid circuit (described above) are selectively associated with peristaltic pumps **202**, **204**, and **206**. The peristaltic pumps articulate with the fluid set of FIG. **3** at the pump segments identified by reference numerals **162**, **166**, **168** and advance the cell suspension or other fluid within the disposable set, as will be understood by those of skill in the art. Apparatus **200** also includes clamps **210**, **212**, **214**, **216**, **218**, **220** and **222**. The clamps are used to control the flow of the cell suspension through different segments of the disposable set, as described above.

(29) Apparatus **200** also includes several sensors to measure various conditions. The output of the sensors is utilized by device **200** to operate one or more wash or processing cycles. One or more pressure transducer sensor(s) **226** may be provided on apparatus **200** and may be associated with a disposable set **100** at certain points to monitor the pressure during a procedure. Pressure transducer **226** may be integrated into an in-line pressure monitoring site (at, for example, tubing segment **136**), to monitor pressure inside separator **101**. Air detector sensor **238** may also be associated with the disposable set **100**, as necessary. Air detector **238** is optional and may be provided to detect the location of fluid/air interfaces.

(30) Apparatus **200** includes weight scales **240**, **242**, **244**, and **246** from which the final product container **150**, waste container **140**, the source container **102** and the in-process container **122**, respectively, may depend and be weighed. The weights of the bags are monitored by weight sensors and recorded during a washing or other procedure. From measurements of the weight sensors, the device determines whether each container is empty, partially full, or full and controls the components of apparatus **200**, such as the peristaltic pumps **202**, **204** and **206** and clamps **210**, **212**, **214**, **216**, **218**, **220** and **222**.

(31) Apparatus **200** includes at least one drive unit or “spinner” **248**, which causes the indirect driving of the spinning membrane separator **101**. Spinner **248** may consist of a drive motor connected and operated by apparatus **200**, coupled to turn an annular magnetic drive member including at least a pair of permanent magnets. As the annular drive member is rotated, magnetic attraction between corresponding magnets within the housing of the spinning membrane separator cause the spinner within the housing of the spinning membrane separator to rotate.

(32) Turning to FIG. **5**, a spinning membrane separation device, generally designated **101**, is shown. Such a device **101** forms part of the disposable circuit **100**.

(33) Device **101** includes a generally cylindrical housing **12**, mounted concentrically about a longitudinal vertical central axis. An internal member **14** is mounted concentric with the central axis **11**. Housing **12** and internal member **14** are relatively rotatable. In the preferred embodiment, as illustrated, housing **12** is stationary and internal member **14** is a rotating spinner that is rotatable concentrically within cylindrical housing **12**, as shown by the thick arrow in FIG. **5**. The boundaries of the flow path are generally defined by gap **16** between the interior surface of housing **12** and the exterior surface of rotary spinner **14**. The spacing between the housing and the spinner is sometimes referred to as the shear gap. The shear gap may be approximately 0.02-0.06 inches (0.05-0.15 cm) and may be of a uniform dimension along axis **11**, for example, where the axis of the spinner and housing are coincident. The shear gap may also vary circumferentially for example,

where the axis of the housing and spinner are offset.

(34) The shear gap also may vary along the axial direction, for example preferably an increasing gap width in the direction. Such a gap width may range from about 0.02 to about 0.075 inches (0.05-0.19 cm). The gap width could be varied by varying the outer diameter of the rotor and/or the inner diameter of the facing housing surface. The gap width could change linearly or stepwise or in some other manner as may be desired. In any event, the width dimension of the gap is preferably selected so that at the desired relative rotational speed, Taylor-Couette flow, such as Taylor vortices, are created in the gap.

(35) Biological fluid is fed from an inlet conduit **20** through an inlet orifice **22**, which directs the fluid into the fluid flow entrance region in a path tangential to the circumference about the upper end of the spinner **14**. At the bottom end of the cylindrical housing **12**, the housing inner wall includes an exit orifice **48**.

(36) Cylindrical housing **12** is completed by a bottom end housing terminating in an outlet orifice **46** concentric with the central axis.

(37) In the illustrated embodiment, the surface of the rotary spinner **14** is at least partially, and is preferably substantially or entirely, covered by a cylindrical porous membrane **62**. The membrane **62** may have a nominal pore size between 0.8 and 10 microns (μm), for example. Membranes may be fibrous mesh membranes, cast membranes, track-etched membranes or other types of membranes that will be known to those of skill in the art. For example, in one embodiment, the membrane may have a polyester mesh (substrate) with nylon particles solidified thereon, thereby creating a tortuous path through which only certain sized components will pass. In an embodiment, the nylon membrane may have a pore size of approximately $0.8\ \mu\text{m}$ and a thickness of approximately $150\ \mu\text{m}$ or greater. Membranes of this type will typically retain all cellular components (e.g., red blood cells, white blood cells) and certain formed blood components, e.g., platelets. In another embodiment, the membrane may be made of a thin (approximately $10\ \mu\text{m}$ thick) sheet of unsupported polycarbonate, for example, with a pore size of approximately $4.0\ \mu\text{m}$. In this embodiment, pores (holes) may be cylindrical and larger than those described above. The pores may be sized to allow small formed components (e.g., platelets, microparticles, etc.) to pass, while the desired cells (e.g., white blood cells and larger red blood cells) are collected.

(38) Having thus described the processor, including disposable circuit **100** and reusable hardware **200**, reference is made to FIG. **2** to discuss additional details of the control unit or controller **300**. As mentioned above, the controller **300** may include a microprocessor **304** (which, in fact may include multiple physical and/or virtual processors). According to other embodiments, the controller **300** may include one or more electrical circuits designed to carry out the actions described herein. In fact, the controller **300** may include a microprocessor and other circuits or circuitry. In addition, the controller **300** may include one or more memories **306**. The instructions by which the microprocessor **304** is programmed may be stored on the memory **306** associated with the microprocessor **304**, which memory/memories **306** may include one or more tangible non-transitory computer readable memories, having computer executable instructions stored thereon, which when executed by the microprocessor **304**, may cause the microprocessors **304** to carry out one or more actions as described below.

(39) As is also illustrated in FIG. **2**, the controller **300** may be coupled to one or more of the structures described above, for example to receive information (e.g., in the form of signals) from these structures or to provide commands (e.g., in the form of signals) to these structures to control the operation of the structures. As illustrated, the controller **300** may be coupled to the scales **240**, **242**, **244**, **246**, the sensors **226**, **238** and the at least one input **302** to receive information from those devices. Additionally, the controller **300** may be coupled to the pumps **202**, **204**, **206**, the clamps **210**, **212**, **214**, **216**, **218**, **220**, **222**, and the drive **248** to provide commands to those devices to control their operation. It may also be possible that the controller **300** receives information from and provides commands to a given structure, such as one of the structures already mentioned. The

controller **300** may be directly electrically connected to these structures to be coupled to them, or the controller **300** may be directly connected to other intermediate equipment that is directly connected to these structures to be coupled to them.

(40) The at least one input **302** may include a number of different devices according to the embodiments described herein. For example, the input **302** could include a keyboard or keypad by which a user may provide information and/or instructions to the controller **300**. Alternatively, the input **302** may be a touch screen, such as may be used in conjunction with a video display **308** that is disposed on the front panel **201** of the device **200**, the video display **308** also being coupled to the controller **300**. The input could also include a reader or scanner, such as a barcode reader or scanner or an RFID reader. The assembly of the input/touch screen **302** and video display **308** may be one of the afore-mentioned structures to which the controller **300** is coupled from which the controller **300** receives information and to which the controller **300** provides commands. According to still other embodiments, the input **302** may be in the form of computer equipment that permits the cell processing system including the controller **300** to communicate (whether via wires, cables, etc. or wirelessly) with other cell processing systems over a local network, or with other cell processing systems or other computer equipment (e.g., a server) over local networks, wide area networks, or the Internet. According to such an embodiment, the input may include an internal transmitter/receiver device.

(41) Having discussed the structure of embodiments of the cell processing system disclosed herein, the operation of the cell processing system is now discussed. In this regard, reference is made to U.S. Patent Application Pub. No. US 2013/0092630, the contents of which are incorporated herein by reference, which document discloses methods and systems for washing biological cells using a reusable hardware apparatus and disposable fluid circuit including a spinning membrane separator which may be generally applicable to the cell processing system described herein. The methods disclosed in this document involve the processing of biological cells, such as mononuclear cells for subsequent therapeutic administration.

(42) In general terms, the operator may first activate (e.g., switch on) apparatus **200**, at which point the apparatus **200** conducts self-calibration checks, including the checking of the peristaltic pumps **202**, **204**, **206**, clamps **210**, **212**, **214**, **216**, **218**, **220**, **222**, and sensors **226**, **238**. Apparatus **200** may then prompt the user to enter or modify process parameters using the input **302**, including by way of example and not by way of limitation the amount of cell suspension to be processed, the number of cycles to take place, etc. The apparatus **200** may then prompt the operator to mount the disposable set **100**, after which apparatus **200** automatically checks to determine whether the disposable set **100** is properly installed. Once the set **100** is properly installed, the controller **300** prompts the operator to connect the biological fluid (e.g., **102** of FIG. 3) via a spike connector or sterile connection (e.g., **103**, **104** of FIG. 3) and the wash medium (e.g., **135a**, **135b** of FIG. 3) via a spike connector (e.g., **134a**, **134b** of FIG. 3). In one embodiment, the biological fluid/cells may be apheresis-collected mononuclear cells, and the wash medium may be a saline solution.

(43) Once the operator confirms that the solutions are connected, the controller **300** primes the disposable set **100**. In the embodiment discussed above, the set **100** may be primed with saline, although other bio-compatible aqueous solutions may also be used. The controller **300** then commences processing the biological fluid/cells. The biological fluid/cells is/are transferred from source container (e.g., **102** of FIG. 3) through the set to the spinning membrane separator **101** via the operation of one or more peristaltic pumps **202**, **204** and **206**. In a similar fashion, the wash medium is delivered from its container (e.g., **135a**, **135b** of FIG. 3) through the set to the spinning membrane separator **101**. The biological cells are collected in either an in-process bag (e.g., **122** of FIG. 3) for additional processing or in a product container (e.g., **150** of FIG. 3), while supernatant is separated and removed to waste container (e.g., **140** of FIG. 3). Once the processing is completed, the controller prompts the operator to sample, seal and remove the product container **150**.

(44) A specific embodiment of a method **400** of operating the apparatus **200** is provided in FIG. **6**. According to this embodiment, the method **400** of operating the apparatus **200** includes several steps, which steps may be grouped or organized into one or more cycles. For example, reduction, rinse and dilution steps **404**, **406**, **408** may define a first cycle, reduction, rinse, and dilution steps **414**, **415**, **416**, **418** may define an optional intermediate cycle (which cycle may be omitted, or the steps **414**, **415**, **416** and/or **418** may be repeated several times to define intermediate cycles—e.g., a 6-cycle procedure may involve the performance of some or all of steps **414-418** a total of 4 times), and reduction, rinse, and dilution steps **424**, **426**, **428** may define a final cycle. It will be recognized that an apparatus **200** need not perform every step illustrated in FIG. **6**, but an apparatus **200** may operate as illustrated in FIG. **6** according to this disclosure.

(45) Preliminary to the first cycle, the controller **300** may perform an evaluation of the process to be performed by the apparatus **200** at block **390**. According to certain embodiments, the evaluation is conducted using a mathematical model of the processor, as explained in detail below. The inputs for the model may include the process, or procedure, parameters received from the operator, via the input **302**, for example. In the alternative or in addition, the inputs for the model may include process parameters that are stored by the controller **300**, for example in the memory **306**. These stored inputs may be in the form of default inputs that are used unless inputs are received via the input **302**.

(46) According to some embodiments, the mathematical model may include equations representative of the fluid flows from and to the containers **102**, **122**, **140**, **150**, of the other fluid flows within the processor **100**, **200**, and of the operation of the separator **101**. In fact, according to preferred embodiments, the mathematical model is representative of the operation of the processor **100**, **200** as illustrated in FIG. **6**, from a priming step at block **402** to a final dilution step at block **428**. The model may also include steps not illustrated in FIG. **6**, such as an incubation step post the final dilution step at block **428**. According to such embodiments, the controller **300** evaluates the entire process, from priming to final dilution, before the method **400** continues to block **402**.

(47) The results of the preliminary process evaluation at block **390** may be provided, displayed or used by the controller in different ways. For example, the controller **300** may provide or display outputs calculated as a consequence of the evaluation of the entire mathematical model, or only portions thereof, to the operator at block **392**. Such outputs may include the duration of the process (as a whole), the duration of the priming step, the final volume in the waste container, the final volume in the product container, and the volume required in the wash media containers. The outputs may be provided or displayed on the display unit **308**, for example. In addition or in the alternative, the controller **300** may limit or prevent the operation of the processor **100**, **200** according to the process at block **394** if, for example, the process would cause the processor **100**, **200** to exceed the abilities or performance characteristics of the processor **100**, **200**. The controller **300** may require at block **394** that an operator or an administrator (i.e., a user with greater control privileges than an operator) provide an override code (e.g., via the input **302**) to allow the process to be performed. As an additional or alternative possibility, the controller **300** may limit or prevent operation of the processor **100**, **200** at any point during the method **400** (e.g., at block **404**) if measured in-process conditions differ from those calculated during the evaluation of the process. The measurements may involve signals received by the controller **300** from one or more of the scales **240**, **242**, **244**, **246** mentioned above. The interruption of the process may be overridden by an operator or administrator using an override code (e.g., received by the controller **300** via the input **302**) as mentioned above relative to the action at block **394**.

(48) Following this pre-process evaluation, the controller **300** may cause the apparatus **200** to perform the step of priming the set **100** at block **402**. According to this step, wash media from the wash media containers **135a**, **135b** is transferred to the disposable set **100**. Wash media may also be transferred to the source container **102**. In fact, a small amount of wash media may be transferred to each of the other containers **102**, **122**, **140**, **150** to ensure that the containers are connected **102**,

122, 140, 150. To this end, the controller **300** may cause clamps **214, 216, 218, 220, 222** to open to permit the transfer of fluid to the containers **102, 122, 140, 150**.

(49) Once the priming is complete at block **402**, the method **400** continues to block **404**, where the controller **300** causes the apparatus **200** to perform the first cycle reduction step. According to this step, the controller **300** causes the biological fluid from the source container **102** and wash media from the wash media container(s) **135a, 135b** to be transferred to the separator **101**. For example, the controller **300** may open clamps **214, 212** (and/or **210**) and operate pumps **204, 202** to transfer the fluids from the containers **102, 135a** (and/or **135b**) to the separator **101**. The separator **101** (in conjunction with operation of the drive **248** by controller **300**) produces two streams: a first, or retentate, stream that is directed into the in-process container **122**, and a second, or filtrate, stream that is directed into the waste container **140**. For example, the controller **300** may open clamp **218** and operate pump **206** to cause flow into the in-process container **122** (clamp **220** being closed), and may open clamp **222** to permit flow into the container **140**. After the step of block **404** is complete, the controller **300** causes wash media to be passed through the set (i.e., the set is rinsed) and the media is added to the in-process bag **122** at block **406**. This may be achieved, for example, by closing clamps **214, 222**, while leaving clamps **212** (and/or **210**), **218** open and operating pumps **202, 206**. After block **406**, the method **400** proceeds to block **408**, where the controller **300** causes additional wash media to be added to the in-process bag **122**. When block **408** is complete, the method **400** passes from the first cycle to the intermediate cycle.

(50) At optional block **414**, the controller **300** may cause the apparatus **200** to further reduce the fluid in the in-process bag **122** by transferring the fluid to the separator **101** without additional dilution, and passing the supernatant to the waste container **140** while the cells are returned to the in-process bag **122**. For example, the controller **300** opens clamps **216, 218, 222** and operates pumps **204, 206** and drive **248**. The controller **300** may continue to cause the apparatus **200** to perform this step until certain user-defined limits have been satisfied. It is also possible that the controller **300** may skip this optional step entirely while operating according to the method **400**, and proceed instead to step **415**.

(51) At optional block **415**, the controller **300** may cause the apparatus **200** to operate such that the feed into the separator **101** is maintained at a constant packed cell volume (PCV). Because cells are being processed from the in-process container **122**, concentrated, and then directed back to the in-process container **122**, the PCV of the in-process container **122** would continuously increase. To limit or prevent the continuous increase, the controller **300** causes the apparatus **200** to add wash media at increasing rates. As such, the controller may open clamp **212** (and/or **210**) and clamps **216, 218, 222** while operating pumps **202, 204, 206** and drive **248**, for example.

(52) Once block **415** is complete, the controller **300** may cause the apparatus to perform a rinse of the set at block **416** and to add wash media to the in-process bag **122** at block **418**. When block **418** is complete, the method **400** passes from the intermediate cycle to the final cycle.

(53) The final cycle begins with block **424**, where the controller **300** causes the biological fluid from the in-process container **122** and wash media from the wash media containers **135a, 135b** to be transferred to the separator **101**. For example, the controller **300** may open clamps **216, 212** (and/or **210**) and operate pumps **204, 202** to transfer the fluids from the containers **102, 135a** (and/or **135b**) to the separator **101**. Again, the separator **101** produces two streams: a first, or retentate, stream that is directed into the retentate, or product, container **150** (instead of the in-process container **122**), and a second, or filtrate, stream that is directed into the waste container **140**. For example, the controller **300** may open clamp **220** and operate pump **206** to cause flow into the product container **150**, and may open clamp **222** to permit flow into the container **140**. After the step of block **424** is complete, the controller **300** causes wash media to be passed through the set (i.e., the set is rinsed) and the media is added to the product bag **150** at block **426**. This may be achieved, for example, by closing clamps **216, 222**, while leaving clamps **212** (and/or **210**), **220** open and operating pumps **202, 206**. After the block **426**, the method **400** proceeds to block **428**,

where the controller **300** causes wash media to be added to the product bag **150**. When block **428** is complete, the method **400** may continue with other steps, such as incubation, as are desired before the product bag **150** is sampled, sealed and removed from the apparatus **200**.

(54) Having discussed the method **400**, a method **500** of performing the preliminary or pre-process evaluation of the process (i.e., **390** in FIG. **6**) is illustrated in FIG. **7**. The method **500** begins at block **502**, with receipt by the controller **300** of the input configuration, i.e., the process parameters required for the processor **100**, **200** to perform the process or procedure. The controller **300** may receive the input configuration via the input **302**, as mentioned above. As also mentioned above, the controller **300** may receive the process parameters from the memory **306** associated with the controller **300**. In fact, certain process parameters may be initialized to a default value according to values stored in the memory **306**, some or all of which may be modified by the user (e.g., operator or administrator) via the input **302**, for example.

(55) The method **500** may continue at block **504**, where the input configuration received at block **502** is evaluated for completeness. For example, the controller **300** may verify that a value has been received for each process parameter in the input configuration. Further, the controller **300** may verify that the values received fall within a preset range for such parameters. The controller **300** may perform other verifications as well. If the controller **300** determines that one or more of the process parameters are missing from the input configuration (or fail to fall within the required range, for example), the process **500** returns to block **502**. Otherwise, the method **500** continues to block **506**.

(56) At block **506**, the controller **300** performs calculations representative of the priming of the set **100** (see, block **402** of FIG. **6**). For example, information regarding the number of priming actions and the identity of the source of the priming fluid (and whether that source is also used as the source for the wash media for the entire procedure) may be received. Based on this information in combination with information relating to the volume of the tubing, of the separator, etc., the controller **300** may calculate the volumes and volume fractions expected to be present in the containers at the end of the priming step, as well as the time required to perform this part or portion of the procedure. The method then continues to block **508**.

(57) At block **508**, the controller **300** performs the calculations representative of the separation of the biological fluid into two streams. This is representative of the separation occurring at block **404** of FIG. **6**, for example. As part of the calculations, the controller sets flow rates for each of a plurality of volumes (each volume representing one of the containers and the separator). The controller **300** also sets the initial volumes for certain of the containers and initial volume fractions. Based on this information, the controller **300** then calculates the volumes and volume fractions expected to be present in the containers at the end of the separation step, as well as other information, such as the time required to complete this part of the process and/or the time required to complete the process to this point.

(58) In doing so, the controller **300** at block **508** uses the final volumes and final volume fractions from the preceding step as the initial volumes and initial volume fractions for this step. According to certain embodiments, including the illustrated embodiment, the controller **300** generally uses the final volumes and final volume fractions from the preceding step as the initial volumes and initial volume fractions for the following step. As a consequence, to calculate the outputs discussed above, the controller **300** first carries out the calculations for every step of the process or procedure, because each succeeding step builds on each prior step.

(59) A comparison may be performed between certain calculated in-process conditions and controls for those conditions at block **510**. While the comparison is illustrated as a separate block **510**, the comparisons may be made while the calculations that occur as part of block **508** are performed. The comparison may involve determining if the calculated condition matches a control. In this regard, “matching” may include being identical to the control value, or within a certain range of control values. “Matching” may also include satisfying a particular relationship to the control

value, such as exceeding or not exceeding the control value. If the comparison is not satisfied (i.e., the calculated value does not match the control value), then a warning or error indication may be provided to the operator, via the video display **308**, for example.

(60) At blocks **512**, **514**, the controller **300** performs calculations representative of the rinse and dilution actions performed, for example, at blocks **406**, **408** in FIG. **6**. As part of this calculation, the controller **300** receives information as to which wash media volume/container (e.g., container **135a** or **135b**) will provide the wash media for the rinse and/or dilution steps. The controller **300** uses the volumes and volume fractions from the previous step as the initial values, and then calculates the final volumes and volume fractions. The controller **300** may also calculate the procedure time for this part (or step) and/or to the completion of the step.

(61) The method **500** continues at block **516**, **520**, where calculations are performed representative of the actions performed at blocks **414**, **415** in FIG. **6**. At block **520**, because the wash media will be added at increasing rates during the corresponding step of the process **400**, the controller **300** approaches the calculation of the rates, volumes and volume fractions as a series of calculations performed over an iterative timescale. Alternatively, if either corresponding step of the process (i.e., **414**, **415**) is omitted, then the final volumes and volume fractions at blocks **516**, **520** are set equal to the initial volumes and volume fractions, and the time for the step is set equal to zero.

(62) As was the case relative to the calculations performed at block **508**, the method **500** includes comparisons of some of the calculated conditions with controls for those conditions at blocks **518**, **522** (similar to block **510**, above). As was also the case above, while the comparisons are illustrated as separate blocks **518**, **522**, the comparisons may occur during the calculations at blocks **516**, **520**. The method then continues at blocks **524**, **526** with calculations representative of the rinse and dilution steps at blocks **416**, **418** in FIG. **6**. The calculations performed here are similar to those performed at blocks **512**, **514**.

(63) In the same manner that the foregoing calculations and comparisons at blocks **516**, **518**, **520**, **522**, **524**, **526** may be omitted if some or all of the steps of the intermediate cycle (i.e., blocks **414**, **415**, **416**, **418**) are omitted, the calculations and comparisons at blocks **516**, **518**, **520**, **522**, **524**, **526** may be repeated if some or all of the steps of the intermediate cycle are repeated to define a process of more than three cycles.

(64) The method **500** concludes with calculations at blocks **528**, **532**, **534** representative of the actions at blocks **424**, **426**, **418** in FIG. **6**. The method also performs a comparison of calculated conditions and controls at block **530**, similar to the comparison described above at block **510**. The calculations performed at blocks **528**, **532**, **534** and the comparisons performed at block **530** are similar to those described above relative to blocks **508**, **512**, **514** and **510**, and as such will not be repeated. The method **500** then concludes, with the subsequent use of the outputs and/or in-process conditions at blocks **392**, **394** of FIG. **6** as discussed above.

(65) The systems and methods described herein may be effective, for example, in the washing of cells such as red blood cells and/or white blood cells. In one example of red cell washing, stored red blood cells may be washed to remove accumulated free hemoglobin, spent storage solution, or extracellular components. The washing solution may be sterile docked or otherwise included in the closed system of the disposable processing set of the type described above. The treated cells may then be washed with the washing solution such as saline, Adsol or E-Sol (the latter of which are red blood cell storage solutions and generally comprise dextrose, mannitol and a buffer) to reconstitute the red blood cells for subsequent storage and transfusion.

(66) The initial cell feed may be diluted by combining the feed from container **102** with diluent (wash solution) from container **135** at branched connector **126**. In one embodiment, diluent from container **135** may initially be drawn into separator, followed by the cell feed drawn from container **102** and combined with the diluent, as described.

(67) Thus, an improved method and system have been disclosed for the processing of biological cells. The description provided above is intended for illustrative purposes only and is not intended

to limit the scope of the invention to any specific method, system, or apparatus, or device described herein except as may be explicitly delineated above.

(68) In conclusion, according to one aspect, a cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive the at least one process parameter via the input, to calculate at least one output based on the at least one process parameter, and to provide the at least one calculated output to an operator prior to causing the processor to operate according to the process.

(69) The at least one calculated output includes one or more of a time to complete the process, or a part thereof, a volume of wash media consumed, a volume of waste fluid generated, and a volume of a residual remaining in a product. Further, the controller may be configured to calculate the at least output based on the at least one process parameter and each step of the process.

(70) According to another aspect, a cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive the at least one process parameter via the input, to calculate at least one in-process condition based on the at least one process parameter, to compare the at least one calculated in-process condition with at least one in-process condition control, and to provide an error indication to an operator if the at least one calculated in-process condition does not match the at least one in-process condition control prior to causing the processor to operate according to the process.

(71) According to such an aspect, the controller may be configured to calculate a plurality of in-process conditions corresponding to each step that defines the process, to compare a set of the plurality of calculated in-process conditions with a set of in-process condition controls, and to provide an error indication to the operator if any element of the set of calculated in-process conditions does not match a corresponding element of the set of in-process condition controls.

(72) According to a still further aspect, a cell processing system includes a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams according to a process including at least one process parameter, and a controller coupled to the processor and an input. The controller is configured to receive at least one process parameter via the input, to calculate at least one in-process condition based on the at least one process parameter prior to causing the processor to operate according to the process, to subsequently cause the processor to operate according to the process, to measure at least one in-process condition during operation of the processor, to compare the at least one calculated in-process condition with at least one measured in-process condition, and to provide an error indication to an operator if the at least one measured in-process condition does not match the at least one calculated in-process condition.

(73) According to such an aspect, the controller may be configured to calculate a plurality of in-process conditions corresponding to each step that defines the process prior to causing the processor to operate according to the process, to compare a set of the plurality of calculated in-process conditions with a set of measured in-process conditions, and to provide an error indication to the operator if any element of the set of measured in-process conditions does not match a corresponding element of the set of calculated in-process conditions.

(74) According to any of the foregoing aspects, the processor may include a disposable fluid circuit and reusable hardware. In fact, the disposable fluid circuit may include a spinning membrane separation device, at least one container, and tubing connecting the spinning membrane and the one or more containers.

- (75) According to yet another aspect, a method for evaluating a process is provided, the process to be performed using a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams. The method includes receiving the at least one process parameter to be used during the process, calculating at least one output based on the at least one process parameter, and providing the at least one calculated output to an operator prior to operating the processor according to the process.
- (76) According to such an aspect, the at least one calculated output may be one of a time to complete the process, or a part thereof, a volume of wash media consumed, a volume of waste fluid generated, and a volume of a residual remaining in a product. The step of calculating at least one output may include calculating the at least output based on the at least one process parameter and each step of the process.
- (77) According to a further aspect, a method for evaluating a process is provided, the process to be performed using a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams. The method includes receiving at least one process parameter to be used during the process, calculating at least one in-process condition based on the at least one process parameter, comparing the at least one calculated in-process condition with at least one in-process condition control, and providing an error indication to an operator if the at least one calculated in-process condition does not match the at least one in-process condition control prior to operating the processor according to the process.
- (78) The method may further include calculating a plurality of in-process conditions corresponding to each step that defines the process, comparing a set of the plurality of calculated in-process conditions with a set of in-process condition controls, and providing an error indication to the operator if any element of the set of calculated in-process conditions does not match a corresponding element of the set of in-process condition controls.
- (79) According to a still further aspect, a method for evaluating a process is provided, the process to be performed using a processor connectable to a source container filled with a biological fluid, the processor including a separator configured to separate the biological fluid from the source container into at least two streams. The method includes receiving at least one process parameter to be used during the process, calculating at least one in-process condition based on the at least one process parameter prior to operating the processor according to the process, causing subsequently the processor to operate according to the process, measuring at least one in-process condition during operation of the processor, comparing the at least one calculated in-process condition with at least one measured in-process condition, and providing an error indication to an operator if the at least one measured in-process condition does not match the at least one calculated in-process condition.
- (80) Furthermore, the method according to this aspect may include calculating a plurality of in-process conditions corresponding to each step that defines the process prior to causing the processor to operate according to the process, comparing a set of the plurality of calculated in-process conditions with a set of measured in-process conditions, and providing an error indication to the operator if any element of the set of measured in-process conditions does not match a corresponding element of the set of calculated in-process conditions.

Claims

1. A blood processing system comprising: a blood processing apparatus comprising reusable hardware and a disposable fluid circuit, the blood processing apparatus comprising a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component in a product container, the reusable hardware comprising a weight scale

configured to weigh the product container and an air detector sensor; a touch screen configured to receive input from an operator; a transmitter/receiver device configured to communicate over a network and to receive process parameters from a server computer; and a processing circuit coupled to the blood processing apparatus and the network, the processing circuit configured to receive the process parameters from the server computer and store them in a memory, the processing circuit configured to store default process parameters in the memory, the processing circuit configured to: receive an activation from the operator to switch on the blood processing system; conduct self-checks on the blood processing apparatus; receive input data for a value of a process parameter from the touch screen; verify that the value is within a preset range for the value; prior to operating the blood processing apparatus according to a centrifugal separation process, store a pre-process calculation of a volume of the blood component to be collected in the product container and provide an indication of the volume to an operator via the touch screen; prompt the operator to mount the disposable fluid circuit; automatically check to determine whether the disposable fluid circuit is installed; prime tubing of the disposable fluid circuit with a fluid; commence the centrifugal separation process on the biological fluid; check the air detector sensor for a fluid/air interface; weigh the product container during the centrifugal separation process; process the biological fluid until the volume of the blood component to be collected is collected; and upon completion of the centrifugal separation process, prompt the operator.

2. The blood processing system of claim 1, further comprising a barcode reader, wherein the processing circuit is further configured to receive input data for a value for a second process parameter from a barcode reader.

3. The blood processing system of claim 1, wherein the disposable fluid circuit comprises a saline access site.

4. The blood processing system of claim 1, wherein the blood component to be collected is blood cells.

5. A blood processing system comprising: a blood processing apparatus comprising reusable hardware and a disposable fluid circuit, the blood processing apparatus comprising a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component in a product container, the reusable hardware comprising a weight scale configured to weigh the product container and an air detector sensor; a touch screen configured to receive input from an operator; a transmitter/receiver device configured to communicate over a network and to receive data from a server computer; and a processing circuit coupled to the blood processing apparatus and the network, the processing circuit configured to receive the data from the server computer and store the data in a memory, the processing circuit configured to store default process parameters in the memory, the processing circuit configured to: receive an instruction from the operator to activate the blood processing system; conduct self-checks on the blood processing apparatus; receive input data for a value of a process parameter from the touch screen; verify that the value is within a preset range for the value; prior to operating the blood processing apparatus according to a centrifugal separation process, store a pre-process calculation of a volume of the blood component to be collected in the product container based at least in part on the value of the process parameter received from the touch screen; provide an indication of the volume to be collected to an operator via the touch screen; prompt the operator to mount the disposable fluid circuit; automatically check to determine whether the disposable fluid circuit is installed; prime tubing of the disposable fluid circuit with a fluid; commence the centrifugal separation process on the biological fluid; check the air detector sensor for a fluid/air interface; weigh the product container during the centrifugal separation process; process the biological fluid until the volume of the blood component to be collected is collected.

6. The blood processing system of claim 5, wherein the processing circuit is further configured to, upon completion of the centrifugal separation process, prompt the operator.

7. The blood processing system of claim 5, wherein the pre-process calculation is based at least in

part on one of the default process parameters stored in the memory.

8. The blood processing system of claim 5, wherein the pre-process calculation is based at least in part on the data received from the server computer.

9. The blood processing system of claim 5, further comprising a barcode reader, wherein the processing circuit is further configured to receive input data for a value for a second process parameter from a barcode reader.

10. The blood processing system of claim 5, wherein the blood component to be collected is blood cells.

11. A blood processing system comprising: a blood processing apparatus comprising reusable hardware and a disposable fluid circuit, the blood processing apparatus comprising a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component to a product container, the reusable hardware comprising a weight scale configured to weigh the product container and an air detector sensor; a touch screen configured to receive input from an operator; and a control unit coupled to the blood processing apparatus, the control unit configured to store default process parameters in the memory, the control unit configured to: receive an instruction from the operator to activate the blood processing system; conduct self-checks on the blood processing apparatus; receive input data for a value of a process parameter from the touch screen; prior to operating the blood processing apparatus according to a centrifugal separation process, calculate a volume of the blood component to be collected in the product container based at least in part on the value of the process parameter received from the touch screen; provide an indication of the volume to be collected to an operator via the touch screen; prime tubing of the disposable fluid circuit with a fluid; commence the centrifugal separation process on the biological fluid using the default process parameters; check the air detector sensor for a fluid/air interface; weigh the product container during the centrifugal separation process; process the biological fluid to collect the volume of the blood component to be collected.

12. The blood processing system of claim 11, wherein the control unit is further configured to verify that the value is within a preset range for the value.

13. The blood processing system of claim 11, further comprising a transmitter/receiver device configured to communicate over a network and to receive data from a server computer, the control unit configured to receive the data from the server computer and store the data in a memory.

14. The blood processing system of claim 13, wherein the control unit is configured to calculate the volume of the blood component to be collected in the product container based at least in part on the data received from the server computer.

15. The blood processing system of claim 14, wherein the control unit is further configured to: prompt the operator to mount the disposable fluid circuit; and automatically check to determine whether the disposable fluid circuit is installed.

16. A blood processing system comprising: a blood processing apparatus comprising reusable hardware and a disposable fluid circuit, the blood processing apparatus comprising a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component in a product container, the reusable hardware comprising a weight scale configured to weigh the product container; a transmitter/receiver device configured to communicate over a network and to receive instructions and/or information from a server computer; a barcode reader configured to receive input data for a pre-process calculation from a barcode; and a processing circuit coupled to the blood processing apparatus and the network, the processing circuit configured to receive the input data for the pre-process calculation from the barcode reader and store the input data in a memory, the processing circuit configured to store default process parameters in the memory, the processing circuit configured to: receive an activation from the operator to switch on the blood processing system; conduct self-checks on the blood processing apparatus; receive the input data for the pre-process calculation from the barcode

reader; verify that the value is within a preset range for the value; prior to operating the blood processing apparatus according to a centrifugal separation process, store a pre-process calculation of a volume of the blood component to be collected in the product container and provide an indication of the volume to an operator via the touch screen; prompt the operator to mount the disposable fluid circuit; automatically check to determine whether the disposable fluid circuit is installed; prime tubing of the disposable fluid circuit with a fluid; commence the centrifugal separation process on the biological fluid; weigh the product container during the centrifugal separation process; process the biological fluid until the volume of the blood component to be collected is collected; and upon completion of the centrifugal separation process, prompt the operator.

17. The blood processing system of claim 16, wherein the blood processing apparatus further comprises an air detector which is associated with the fluid flow circuit and the processing circuit is further configured to check an air detector sensor for a fluid/air interface.

18. The blood processing system of claim 16, wherein the disposable fluid circuit comprises a saline access site.

19. The blood processing system of claim 16, wherein the blood component to be collected is blood cells.

20. A blood processing system comprising: a blood processing apparatus comprising reusable hardware and a disposable fluid circuit, the blood processing apparatus comprising a separator configured to perform a centrifugal separation process on a biological fluid from a source to provide a blood component in a product container, the reusable hardware comprising a weight scale configured to weigh the product container; a touch screen configured to receive input from an operator; a transmitter/receiver device configured to communicate over a network and to receive data from a server computer; and a processing circuit coupled to the blood processing apparatus and the network, the processing circuit configured to receive the data from the server computer and store the data in a memory, the processing circuit configured to store default process parameters in the memory, the processing circuit configured to: receive an instruction from the operator to activate the blood processing system; conduct self-checks on the blood processing apparatus; receive input data for a value of a process parameter from the touch screen; verify that the value is within a preset range for the value; prior to operating the blood processing apparatus according to the centrifugal separation process, store a pre-process calculation of a volume of the blood component to be collected in the product container based at least in part on the value of the process parameter received from the touch screen and/or based on the data received from the server computer; provide an indication of the volume to be collected to an operator via the touch screen; prompt the operator to mount the disposable fluid circuit; prime tubing of the disposable fluid circuit with a fluid; commence the centrifugal separation process on the biological fluid, wherein the centrifugal separation process is based at least in part on the default process parameters; process the biological fluid until the volume of the blood component to be collected is collected.

21. The blood processing system of claim 20, wherein the processing circuit is further configured to, upon completion of the centrifugal separation process, prompt the operator.

22. The blood processing system of claim 20, wherein the pre-process calculation is based at least in part on one of the default process parameters stored in the memory.

23. The blood processing system of claim 20, wherein the processing circuit is configured to automatically check to determine whether the disposable fluid circuit is installed.

24. The blood processing system of claim 20, further comprising a barcode reader, wherein the processing circuit is further configured to receive input data for a value for a second process parameter from a barcode reader.

25. The blood processing system of claim 20, wherein the blood component to be collected is blood cells.
